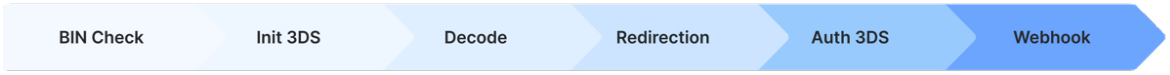
3DS Implementation



Steps

1. BIN Check
2. Init 3DS
3. "threeDSHtmlContent" Decode
4. Redirection
5. Auth 3DS
6. Webhook

Top View

- BIN Check; section of identifying issuer details where API returns information about installment options and card details.
- Init 3DS; first `POST` request of managing two-step 3DS transactions.
- "threeDSHtmlContent"; base64 encrypted HTML page where your end-user expected to validate 3DS page OTP verification.
- Redirection; Following up consumers' payment, iyzico triggers correlated IPN(Instant Payment Notification) to given callbackUrl.
- Auth 3DS; engage step to officialize transaction between Acquirer and Issuer.
- Webhook; real-time notification of relevant payment.

Sample Implementation

3DS implementation consists of 2 subsequent `POST` requests in short.

- Init 3DS
- Auth 3DS

Suggested integration tracks following order;

1. BIN Check

Whether 3DS or NON-3DS payment method, It is advised to identify issuer card feature beforehand. BIN detail services provides further details of card type, card scheme, card association which indicates installment and 3DS features.

Sample Retrieve Installment Request;

```
{
  "price":"100.0",
  "binNumber":"535805"
}
```

Sample Retrieve Installment Response;

```
{
    "status": "success",
    "locale": "tr",
    "systemTime": 1685905139724,
    "installmentDetails": [
        {
            "binNumber": "535805",
            "price": 100.0,
            "cardType": "DEBIT_CARD",
            "cardAssociation": "MASTER_CARD",
            "cardFamilyName": "iyzico DC",
            "force3ds": 0,
            "bankCode": 864,
            "bankName": "iyzico",
            "forceCvc": 0,
            "commercial": 0,
            "dccEnabled": 0,
            "installmentPrices": [
                {
                    "installmentPrice": 100.0,
                    "totalPrice": 100.0,
                    "installmentNumber": 1
                }
            ]
        }
    ]
}
```


2. Init 3DS

3DS payment journey begins with Init 3DS `POST` request.

Sample Init 3DS Request;

```
{
    "locale": "en",
    "price": "3.2",
    "paidPrice": "3.2",
    "installment": 1,
    "paymentChannel": "WEB",
    "basketId": "B67832",
    "paymentGroup": "PRODUCT",
    "paymentCard": {
        "cardHolderName": "Dev iyzico",
        "cardNumber": "5526080000000006",
        "expireYear": "2023",
        "expireMonth": "11",
        "cvc": "200"
    },
    "buyer": {
        "id": "BY789",
        "name": "John",
        "surname": "Doe",
        "identityNumber": "74300864791",
        "email": "email@email.com",
        "gsmNumber": "+905350000000",
        "registrationDate": "2013-04-21 15:12:09",
        "lastLoginDate": "2015-10-05 12:43:35",
        "registrationAddress": "Nidakule Göztepe, Merdivenköy M
        "city": "Istanbul",
        "country": "Turkey",
        "zipCode": "34732",
        "ip": "85.34.78.112"
    },
    "conversationId": "deviyzico",
    "shippingAddress": {
        "address": "Nidakule Göztepe, Merdivenköy Mah. Bora Sok
        "zipCode": "34742",
        "contactName": "Jane Doe",
        "city": "Istanbul",
        "country": "Turkey"
    },
    "billingAddress": {
        "address": "Nidakule Göztepe, Merdivenköy Mah. Bora Sok
        "zipCode": "34742",
        "contactName": "Jane Doe",
        "city": "Istanbul",
        "country": "Turkey"
    },
    "basketItems": [
        {
```

```
            {
                "id": "BI101",
                "price": "1.1",
                "name": "Binocular",
                "category1": "Collectibles",
                "category2": "Accessories",
                "itemType": "PHYSICAL"
            },
            {
                "id": "BI1012",
                "price": "2.1",
                "name": "Binocular",
                "category1": "Collectibles",
                "category2": "Accessories",
                "itemType": "PHYSICAL"
            }
        ],
        "currency": "TRY",
        "callbackUrl": "https://deviyzico.com/"
    }n
```

In return It is expected to receive " `threeDSHtmlContent` " key-value with other pieces;

Sample Init 3DS Response;

```
{
    "status": "success",
    "locale": "en",
    "systemTime": 1685539494559,
    "conversationId": "deviyzico",
    "threeDSHtmlContent":
"PCFkb2N0eXBlIGh0bWw+CjxodG1sIGxhbmc9ImVuIj4KPGhlYWQ+CiAgICA8d
Gl0bGU+aXl6aWNvIE1vY2sgM0QtU2VjdXJlIFByb2Nlc3NpbmcgUGFnZTwvdGl
0bGU+CjwvaGVhZD4KPGJvZHk+Cjxmb3JtIGlkPSJpeXppY28tM2RzLWZvcm0iI
GFjdGlvbj0iaHR0cHM6Ly9zYW5kYm94LWFwaS5peXppcGF5LmNvbS9wYXltZW5
0L21vY2svaW5pdDNkcyIgbWV0aG9kPSJwb3N0Ij4KICAgIDxpbnB1dCB0eXBlP
SJoaWRkZW4iIG5hbWU9Im9yZGVySWQiIHZhbHVlPSJtb2NrNDYtNjg3NjU1ODA
wODI2MjM5NWl5emlvcmQiPgogICAgPGlucHV0IG5hbWU9InRva2VuIiBpZGRlbiIgbmFtZ
T0iYmluIiB2YWx1ZT0iNTUyNjA4Ij4KICAgIDxpbnB1dCB0eXBlPSJoaWRkZW4
iIG5hbWU9InN1Y2Nlc3NVcmwiIHZhbHVlPSJodHRwczovL3NhbmRib3gtYXBpL
ml5emlwYXkuY29tL3BheW1lbnQvaXl6aXBvcy9jYWxsYmFjazNkcy9zdWNjZXN
zLzI3Ij4KICAgIDxpbnB1dCB0eXBlPSJoaWRkZW4iIG5hbWU9ImZhaWx1cmVVc
mwiIHZhbHVlPSJodHRwczovL3NhbmRib3gtYXBpLml5emlwYXkuY29tL3BheW1
lbnQvaXl6aXBvcy9jYWxsYmFjazNkcy9mYWlsdXJlLzI3Ij4KICAgIDxpbnB1d
CB0eXBlPSJoaWRkZW4iIG5hbWU9ImNvbmZpcm1hdGlvblVybCIgdmFsdWU9Imh
0dHBzOi8vc2FuZGJveC1hcGkuaXl6aXBheS5jb20vcGF5bWVudC9tb2NrL2Nvb
mZpcm0zZHMiPgogICAgPGlucHV0IG5hbWU9InRva2VuIiBpZGRlbiIgbmFtZT0iUGFSZXE
iIHZhbHVlPSJjYWIxNjA0YS02MWJiLTQ0NGQtOGExNS1kZDZmMzhjZGRiNjMiP
go8L2Zvcm0+CjxzY3JpcHQgdHlwZT0idGV4dC9qYXZhc2NyaXB0Ij4KICAgIGR
vY3VtZW50LmdldEVsZW1lbnRCeUlkKCJpeXppY28tM2RzLWZvcm0iKS5zdWJta
XQoKTsKPC9zY3JpcHQ+CjwvYm9keT4KPC9odG1sPg=="
}
```


3. "threeDSHtmlContent" Decode

" `threeDSHtmlContent` " represents encrypted 3DS form itself.

Once we render encoded " `threeDSHtmlContent` " parameter from Init 3DS response on the browser, 3DS screen will display in front of the consumer.

Decoded "threeDSHtmlContent" Sandbox Sample;

```
"threeDSHtmlContent":
"PCFkb2N0eXBlIGh0bWw+CjxodG1sIGxhbmc9ImVuIj4KPGhlYWQ+CiAgICA8dGl0b
GU+aXl6aWNvIE1vY2sgM0QtU2VjdXJlIFByb2Nlc3NpbmcgUGFnZTwvdGl0bGU+Cjw
vaGVhZD4KPGJvZHk+Cjxmb3JtIGlkPSJpeXppY28tM2RzLWZvcm0iIGFjdGlvbj0ia
HR0cHM6Ly9zYW5kYm94LWFwaS5peXppcGF5LmNvbS9wYXltZW50L21vY2svaW5pdDN
kcyIgbWV0aG9kPSJwb3N0Ij4KICAgIDxpbnB1dCB0eXBlPSJoaWRkZW4iIG5hbWU9I
m9yZGVySWQiIHZhbHVlPSJtb2NrNDYtNjg3NjU1ODAwODI2MjM5NWl5emlvcmQiPgo
gICAgPGlucHV0IHR5cGU9ImhpZGRlbiIgbmFtZT0iYmluIiB2YWx1ZT0iNTUyNjA4I
j4KICAgIDxpbnB1dCB0eXBlPSJoaWRkZW4iIG5hbWU9InN1Y2Nlc3NVcmwiIHZhbHV
lPSJodHRwczovL3NhbmRib3gtYXBpLml5emlwYXkuY29tL3BheW1lbnQvaXl6aXBvc
y9jYWxsYmFjazNkcy9zdWNjZXNzLzI3Ij4KICAgIDxpbnB1dCB0eXBlPSJoaWRkZW4
iIG5hbWU9ImZhaWx1cmVVcmwiIHZhbHVlPSJodHRwczovL3NhbmRib3gtYXBpLml5e
mlwYXkuY29tL3BheW1lbnQvaXl6aXBvcy9jYWxsYmFjazNkcy9mYWlsdXJlLzI3Ij4
KICAgIDxpbnB1dCB0eXBlPSJoaWRkZW4iIG5hbWU9ImNvbmZpcm1hdGlvblVybCIgd
mFsdWU9Imh0dHBzOi8vc2FuZGJveC1hcGkuaXl6aXBheS5jb20vcGF5bWVudC9tb2N
rL2NvbmZpcm0zZHMiPgogICAgPGlucHV0IHR5cGU9ImhpZGRlbiIgbmFtZT0iUGFSZ
XEiIHZhbHVlPSJjYWIxNjA0YS02MWJiLTQ0NGQtOGExNS1kZDZmMzhjZGRiNjMiPgo
8L2Zvcm0+CjxzY3JpcHQgdHlwZT0idGV4dC9qYXZhc2NyaXB0Ij4KICAgIGRvY3VtZ
W50LmdldEVsZW1lbnRCeUlkKCJpeXppY28tM2RzLWZvcm0iKS5zdWJtaXQoKTsKPC9
zY3JpcHQ+CjwvYm9keT4KPC9odG1sPg=="
```

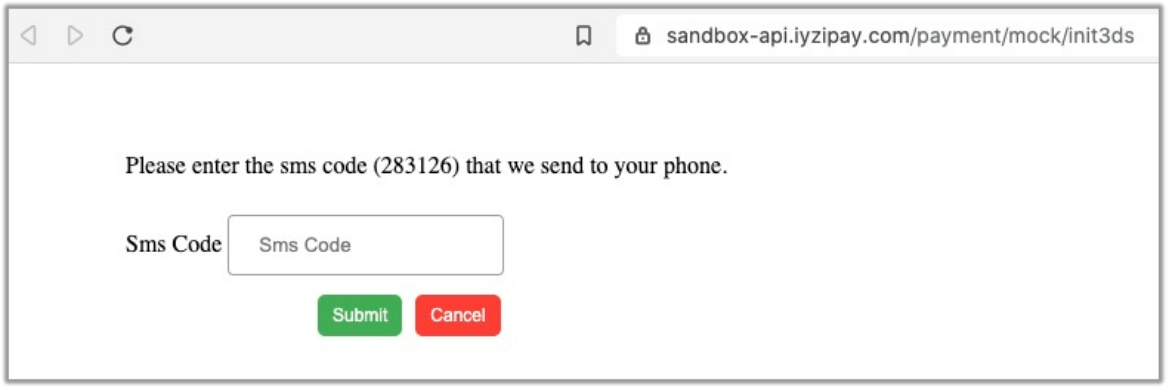

Encoded "threeDSHtmlContent" Sandbox Sample;

```
<!doctype html>
<html lang="en">
<head>
    <title>iyzico Mock 3D-Secure Processing Page</title>
</head>
<body>
<form id="iyzico-3ds-form" action="https://sandbox-api.iyzipay.com/p
    <input type="hidden" name="orderId" value="mock46-6876558008262
    <input type="hidden" name="bin" value="552608">
    <input type="hidden" name="successUrl" value="https://sandbox-ap
    <input type="hidden" name="failureUrl" value="https://sandbox-ap
    <input type="hidden" name="confirmationUrl" value="https://sandb

    <input type="hidden" name="PaReq" value="cab1604a-61bb-444d-8a15
    <input type="hidden" name="termUrl" value="cab1604a-61bb-444d-8a
    <input type="hidden" name="md" value="cab1604a-61bb-444d-8a15-dc

</form>
<script type="text/javascript">
    document.getElementById("iyzico-3ds-form").submit();
</script>
</body>
</html>
```

An example of decoded threeDSHtmlContent interface on the browser;



4. Redirection

Redirection phase ties Init 3DS and Auth 3DS steps overall.

Following up cardholder OTP(One Time Password) confirmation iyzico will automatically redirect the page to the address given in the "callbackUrl" parameter. Redirection `POST` includes;

Input Name	Type	Description
status	String	Service response result (success / failure)
paymentId	String	If verification is successful, iyzico will return a paymentid. It must be set in Auth request
conversationData	String	If verification is successful, iyzico might return. If returns, it must be set in Auth request
conversationId	Long	If set, conversation ID to match request and response
mdStatus	String	**1** for successful payment, **0,2,3,4,5,6,7,8** for failure payments

mdStatuses

" `mdStatus` " interprets merchant plug-in responses on 3DS triangle.

While successful 3DS operations results with `"mdStatus":"1"` on " `callbackUrl",` failure scenarios might have one of the items from list below;

mdStatus	Description
mdStatus = 0	Invalid 3D Secure signature or verification
mdStatus = 2	Card holder or Issuer not registered to 3D Secure network
mdStatus = 3	Issuer is not registered to 3D secure network
mdStatus = 4	Verification is not possible, card holder chosen to register later on system
mdStatus = 5	Verification is not possbile
mdStatus = 6	3D Secure error
mdStatus = 7	System error
mdStatus = 8	Unknown card

5. Auth 3DS

After having completing all the steps above, Its now time to officialize payment operation between Issuer and Acquirer with Auth 3DS request.

Sample Init 3DS Request;

```
{
  "paymentId": "123456789"
}
```

Sample Init 3DS Response;

```
{
    "status": "success",
    "locale": "tr",
    "systemTime": 1685539637584,
    "price": 3.20000000,
    "paidPrice": 3.20000000,
    "installment": 1,
    "paymentId": "123456789",
    "fraudStatus": 1,
    "merchantCommissionRate": 0E-8,
    "merchantCommissionRateAmount": 0E-8,
    "iyziCommissionRateAmount": 0.12800000,
    "iyziCommissionFee": 0.25000000,
    "cardType": "DEBIT_CARD",
    "cardAssociation": "MASTER_CARD",
    "cardFamilyName": "iyzico DC",
    "binNumber": "535805",
    "lastFourDigits": "0006",
    "basketId": "B67832",
    "currency": "TRY",
    "itemTransactions": [
        {
            "itemId": "BI101",
            "paymentTransactionId": "123456789",
            "transactionStatus": 2,
            "price": 1.10000000,
            "paidPrice": 1.10000000,
            "merchantCommissionRate": 0E-8,
            "merchantCommissionRateAmount": 0E-8,
            "iyziCommissionRateAmount": 0.04400000,
            "iyziCommissionFee": 0.08593750,
            "blockageRate": 0E-8,
            "blockageRateAmountMerchant": 0E-8,
            "blockageRateAmountSubMerchant": 0,
            "blockageResolvedDate": "2023-06-08 00:00:00",
            "subMerchantPrice": 0,
            "subMerchantPayoutRate": 0E-8,
            "subMerchantPayoutAmount": 0,
            "merchantPayoutAmount": 0.97006250,
            "convertedPayout": {
                "paidPrice": 1.10000000,
                "iyziCommissionRateAmount": 0.04400000,
                "iyziCommissionFee": 0.08593750,
                "blockageRateAmountMerchant": 0E-8,
                "blockageRateAmountSubMerchant": 0E-8,
                "subMerchantPayoutAmount": 0E-8,
                "merchantPayoutAmount": 0.97006250,
```

```
                    "iyziConversionRate": 0,
                    "iyziConversionRateAmount": 0,
                    "currency": "TRY"
                }
            },
            {
                "itemId": "BI1012",
                "paymentTransactionId": "123456788",
                "transactionStatus": 2,
                "price": 2.10000000,
                "paidPrice": 2.10000000,
                "merchantCommissionRate": 0E-8,
                "merchantCommissionRateAmount": 0E-8,
                "iyziCommissionRateAmount": 0.08400000,
                "iyziCommissionFee": 0.16406250,
                "blockageRate": 0E-8,
                "blockageRateAmountMerchant": 0E-8,
                "blockageRateAmountSubMerchant": 0,
                "blockageResolvedDate": "2023-06-08 00:00:00",
                "subMerchantPrice": 0,
                "subMerchantPayoutRate": 0E-8,
                "subMerchantPayoutAmount": 0,
                "merchantPayoutAmount": 1.85193750,
                "convertedPayout": {
                    "paidPrice": 2.10000000,
                    "iyziCommissionRateAmount": 0.08400000,
                    "iyziCommissionFee": 0.16406250,
                    "blockageRateAmountMerchant": 0E-8,
                    "blockageRateAmountSubMerchant": 0E-8,
                    "subMerchantPayoutAmount": 0E-8,
                    "merchantPayoutAmount": 1.85193750,
                    "iyziConversionRate": 0,
                    "iyziConversionRateAmount": 0,
                    "currency": "TRY"
                }
            }
        ],
        "authCode": "905307",
        "phase": "AUTH",
        "mdStatus": 1,
        "hostReference": "mock00007iyzihostrfn"
    }
```


NOTE : `mdStatus` parameter will only appear on 3DS operations.

6. Webhook

Webhooks are tail subject of our implementation streamline.

Receiving real-time payment notifications leverages overall 3DS experience while ensuring that the notifications are coming from trusted sources, preventing any potential tampering or unauthorized access to sensitive data.

Each payment event triggers webhooks;

- In `15` seconds.
- Until your server responds with `200`.
- For every `10` minutes.
- Max `3` times.

Sample Auth 3DS Webhook

Parametere	Type	Description
iyziEventTime	long	Unix timestamp value of first notification.
iyziEventType	string	Shows the request type. Values: API_AUTH, THREE_DS_AUTH, BKM_AUTH
iyziReferenceCode	string	A unique reference code for the notification
paymentId	long	Unique iyzico reference code of related payment
paymentConversationId	string	Merchant's reference id for the related payment
status	string	Payment status. Values: SUCCESS, FAILURE

Happy path includes `success` parameter in the `status` variable.

Confirmation

To finalize payment operations successfully, correlation of certain variables plays significant role;

- `paymentId`
- `status`
- `price`
- Other Idempotency parameters optionally

Its vital that Auth 3DS response and Webhook notification have to authenticate each other.

- `paymentId` on Auth 3DS response should be equal to `paymentId` from Webhook
- Both of the `status` parameters should be `success`
- `price` parameter amounts should be the same
- And lastly, Idempotency parameters should be the same as well.

If so, congratulations! Just finished 3DS implementation.

Transaction details coudl getfrom merchant panel.

Which Products Can It be Used with?

Online Payment

Marketplace

Subscription

iyzilink

Checkout Form (CF)

PayWithiyzico (PWI)

Advantages

- Faster integration
- Single 3DS method which covers 1.X.X or 2.X.X
- Easy to plug&play among other iyzico APIs

Previous
3DS

Next
Init 3DS

Last updated 4 months ago